Anomaly Detection Statistical Report

Automated Statistical Analysis of Model Predictions

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Total Records Analyzed: 200

Contents

1 Executive Summary

This report presents a comprehensive statistical analysis of 200 anomaly-detection predictions. The dataset contains 177 records classified as normal (88.50%) and 23 records classified as anomalies (11.50%). The mean anomaly score is -0.081661 with a standard deviation of 0.058825. A 95% confidence interval for the mean score was computed as [-0.089864, -0.073459]. A total of 10 statistical outliers were detected using the IQR method.

2 Dataset Statistics

Table 1: Descriptive Statistics of Anomaly Scores

Statistic	**Value**
Count	200
Mean	-0.081661
Median	-0.099957
Variance	0.003460
Standard Deviation	0.058825
Minimum	-0.142160
Maximum	0.122446
Range	0.264607
Q1 (25th percentile)	-0.126831
Q2 / Median (50th percentile)	-0.099957
Q3 (75th percentile)	-0.056129
Interquartile Range (IQR)	0.070702
Skewness	1.405215
Kurtosis	1.482592

2.1 Prediction Class Distribution

Table 2: Class Distribution

Class	**Count**	**Percentage**
Normal (0)	177	88.50%
Anomaly (1)	23	11.50%

3 Confidence Interval Analysis

Using Student's t-distribution with 199 degrees of freedom, the standard error of the mean is 0.004160. The 95% confidence interval for the population mean anomaly score is:

$$CI_{95\%} = [-0.089864,\ -0.073459]$$

with a margin of error of 0.008203. This indicates that we are 95% confident the true mean anomaly score of the underlying population lies within this interval.

4 Visualizations

4.1 Score Distribution

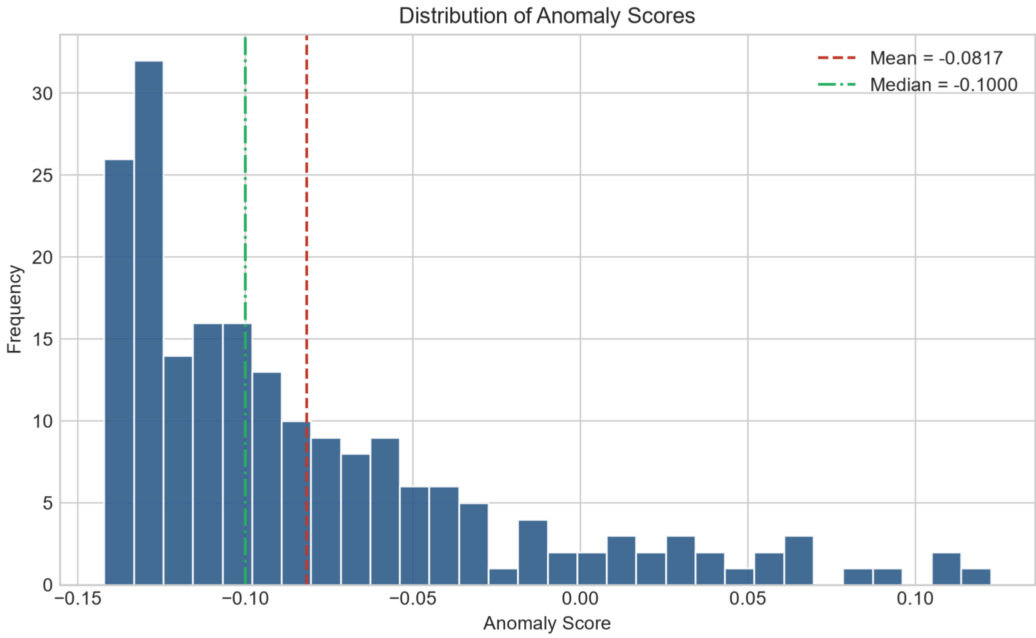

Figure 1: Histogram of anomaly scores with mean and median indicators

4.2 Class Proportion

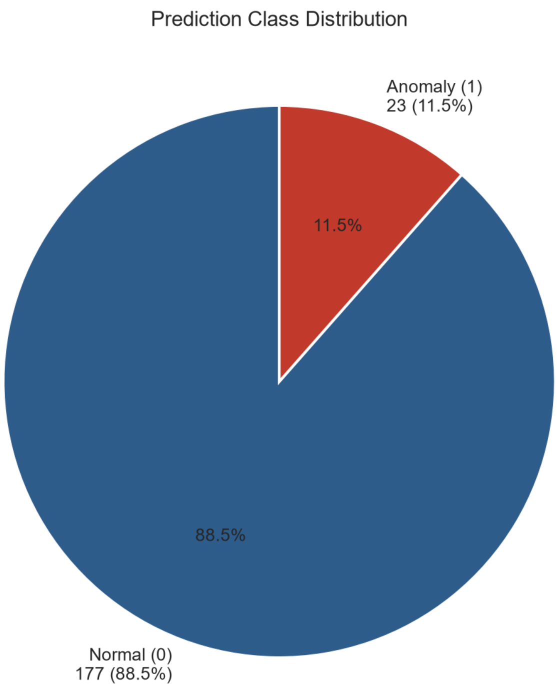

Figure 2: Proportion of normal vs. anomalous predictions

4.3 Score Spread

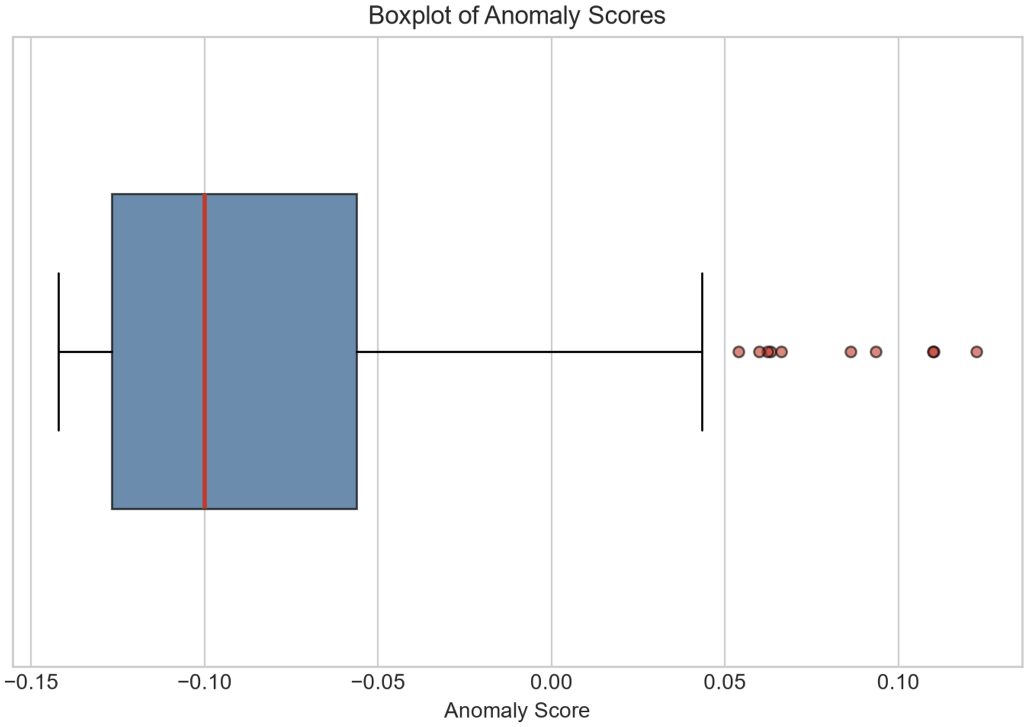

Figure 3: Boxplot showing quartiles and outliers of anomaly scores

5 Top Anomalies

The table below lists the top 20 records with the lowest anomaly scores (most anomalous), color-coded by severity relative to the IQR fences (lower fence = -0.232883, upper fence = 0.049924).

Table 3: Top Anomalies by Score

Row ID	Prediction	Anomaly Score
59	0	-0.142160
113	0	-0.141631
52	0	-0.141369
88	0	-0.141356
148	0	-0.140968
181	0	-0.140376
8	0	-0.140352
142	0	-0.139398
98	0	-0.139116
40	0	-0.138643
30	0	-0.138528
43	0	-0.138480
89	0	-0.137710
12	0	-0.136652
42	0	-0.136531
13	0	-0.136000
152	0	-0.135986
58	0	-0.135702

Row ID	Prediction	Anomaly Score
5	0	-0.135458
31	0	-0.135394